

A parametric study of the turbulent energy transfer in three-dimensional lattice-Boltzmann Hall-MHD plasma simulations

Cite as: Phys. Plasmas **33**, 012304 (2026); doi: [10.1063/5.0298970](https://doi.org/10.1063/5.0298970)

Submitted: 25 August 2025 · Accepted: 30 December 2025 ·

Published Online: 20 January 2026



View Online



Export Citation



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C. L. Vásconez,^{1,a)} R. Foldes,² R. Marino,² R. E. Manobanda,³ E. Lévêque,² D. Telloni,⁴
and L. Sorriso-Valvo^{5,6}

AFFILIATIONS

¹Departamento de Física, Escuela Politécnica Nacional, Quito, Ecuador

²CNRS, École Centrale de Lyon, INSA Lyon, Université Claude Bernard Lyon 1, Laboratoire de Mécanique des Fluides et d'Acoustique, UMR5509, F-69134 Écully, France

³Unidad Técnica Pet-Ciclotrón, Hospital de Especialidades Carlos Andrade Marín, Quito, Ecuador

⁴INAF – Turin Astrophysical Observatory, Via Osservatorio 20, 10025 Pino Torinese (TO), Italy

⁵CNR, Institute for Plasma Science and Technology (ISTP), via Amendola 122/D, 70126 Bari, Italy

⁶Department of Electromagnetism and Plasma Physics, School of Electrical Engineering and Computer Science, KTH Royal Institute of Technology, Teknikringen 31, 11428 Stockholm, Sweden

^{a)}Author to whom correspondence should be addressed: christian.vasconez@epn.edu.ec

ABSTRACT

Observations from heliospheric missions have deepened our understanding of the solar wind. However, the complexity arising from its quasi-collisionless and multiscale nature leaves open several key questions on how energy is transferred and dissipated in the interplanetary plasma. In this parametric study, we investigate turbulent energy transfer and intermittent plasma processes in three-dimensional direct numerical simulations, in the Hall-MHD framework, performed with the fast lattice-Boltzmann algorithm for magnetohydrodynamics experiments [Foldes *et al.*, “Evidence of dual energy transfer driven by magnetic reconnection at subion scales,” Phys. Rev. E **110**, 055207 (2024)]. We analyze properties of the global and pseudo-local energy transfer to characterize the scale-dependent behavior of the energy cascade in Hall-MHD plasmas by means of the so-called local energy transfer proxy (or LET), the analysis of the probability distribution functions of the macroscopic fields, and the flatness of their increments. Our results indicate that scale-dependent structures, small-scale intermittency, and statistical properties in the plasma flow are sensitive to variations of the Hall term, providing insights that are potentially useful for the interpretation of space plasma dynamics.

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I. INTRODUCTION

In situ and remote sensing observations of solar and heliospheric plasmas allowed through the decades for compelling characterizations of turbulence in space, revealing the existence of multiple regimes that make the dynamics of the interplanetary medium extremely complex, holding several open questions.^{2–7} Among others, some of these concern the physical phenomena that initiate and sustain the turbulent energy cascade, the wave–particle interaction mechanisms mediating dissipation and heating in the solar wind,⁸ and the feedback of magnetic reconnection, contributing to the redistribution of the energy at

subion scales.¹ Numerous plasma models (e.g., Adhikari *et al.*,^{9,10} Perrone *et al.*,¹¹ and Jain *et al.*¹²) enabled the study of how energy is transferred in space across fluid and kinetic regimes, often showing good agreement with observations. That happens, for instance, when one explores stochastic ion/proton heating,¹³ collisionless damping,^{14–16} and the acceleration of plasma particles by magnetic reconnection.^{17,18} Dissipation in the solar wind occurs beyond the small-scale end of the MHD inertial range, i.e., between ion/proton and electron scales, and perhaps further beyond. Simulations should therefore resolve plasma dynamics from the MHD down to the kinetic scales, to

be able to perform comprehensive investigations of the energy cascade in plasmas, which is challenging, of course, from a computational standpoint. Kinetic simulations, typically performed using Vlasov or particle-in-cell (PIC) codes, are indeed computationally really intensive, often relying on nonphysical parameter choices, such as artificially reduced ion-to-electron mass ratio, or filters to remove the energy from the system at the small scales. The Hall-MHD model represents a robust fluid-like approximation of the plasma that is able to capture certain effects of the dynamics at sub-ion scales, particularly those associated with the Hall currents, though it does not fully describe the physics of the kinetic scales. Phenomena such as wave-particle resonances, temperature anisotropies, and non-Maxwellian features of particle distribution functions do still require kinetic simulations.¹⁹ Nevertheless, thanks to a reduced complexity, Hall-MHD simulations represent, in many cases, a very good compromise to get insights into such plasma dynamics as those developing in outer space.

Leveraging these arguments, here we propose fundamental investigations of plasma turbulence features in the Hall-MHD framework. In particular, in the formulation of the Hall-MHD governing equations we consider, the gradient of the electron pressure is neglected in the induction equation, a simplification commonly adopted under the assumption of isothermal or barotropic electron behavior. In this work, we analyze global and pseudo-local properties of the energy transfer in simulations of three-dimensional (3D) turbulent plasmas, aiming at assessing the contribution of large-scale turbulent dynamics to phenomena developing at the smaller kinetic scales.^{15,20}

In incompressible magnetohydrodynamic turbulence, statistics of fluctuations can be characterized using power spectra, intermittency properties, and scaling laws.^{21,22} In the inertial range, typical power laws are observed, whose exponents can be predicted by phenomenological considerations. The MHD turbulence phenomenology by Iroshnikov-Kraichnan predicts a spectral exponent of $-3/2$, slightly shallower than the exponent predicted by Kolmogorov for the fluid case, $-5/3$,^{23,24} the solar wind observations of magnetic spectra are compatible, in general, with one of the two phenomenological expectations, depending on the level of evolution of the turbulence in space.^{21,25,26} Similar to the hydrodynamic case, scaling relations can be obtained for the mixed third-order longitudinal structure functions of velocity and magnetic field in Hall-magnetohydrodynamic plasmas. Such relations have been studied in depth since the original derivation for the magnetohydrodynamic case by Politano and Pouquet,^{23,27–29} and assessed in numerous solar wind studies (e.g., Sorriso-Valvo *et al.*,³⁰ Galtier,³¹ Marino *et al.*,^{32,33} Bandyopadhyay *et al.*^{34,35}) and numerical simulations (e.g., Verdini *et al.*,³⁶ Vázquez *et al.*³⁷). Intermittency is a defining feature of fully developed turbulence in both fluids and plasmas.³⁸ Observations from experiments and numerical simulations show that dissipation occurs through intense, localized bursts. This behavior is widely observed in space plasmas³⁹ and geophysical fluids, where non-stationary energetic events also emerge at scales comparable to the mean flow.^{40–43} Shape of spectra, intermittency properties, and features of the third-order scaling relations built on the macroscopic fields are all interconnected; however, each highlights specific aspects of the field fluctuations in turbulent flows.

Here, we present a parametric study of turbulent plasmas obtained using Hall-MHD simulations performed using the fast lattice-Boltzmann (LB) algorithm for magnetohydrodynamics experiments - FLAME.⁴⁴ Lattice-Boltzmann methods adopt a

mesoscopic representation, where the macroscopic flow behavior emerges from the repeated application of simple collision and streaming rules to populations of particles moving along the links of a regular cubic lattice. This structure makes LB codes, especially well-suited for execution on multi-processor clusters, including systems equipped with multiple GPUs. We comparatively examine power spectral densities, intermittency, and third-order scaling relations of the simulated plasma flows to describe the turbulence properties under different conditions. Furthermore, we explore possible differences across the runs in the locality of energy dissipation, by computing a heuristic proxy of the scale-dependent local energy transfer rate (LET),⁴⁵ obtained using ensemble averages over spatial increments.

The manuscript is structured as follows: in Sec. II, we present the numerical implementation of the Hall-MHD model utilized to simulate the plasma. A characterization of the turbulent transfer in the Hall-MHD runs performed is then proposed in Sec. III, using a surrogate of the Politano-Pouquet law. In Sec. IV, we estimate the LET in all the simulations, while visualizations are only shown for RUN III, which exhibits the strongest turbulence level. We finally summarize our results in Sec. V.

II. HALL-MHD PLASMA

A novel aspect of the research proposed here is that the parametric study designed to get insights on energy transfer and intermittency in plasmas, as observed in space,^{21,22} is based on Hall-MHD simulations performed with the LB code FLAME. Incompressibility cannot be strictly enforced in LB methods, as it would require particles moving at infinite speed along the lattice links. However, in the LB approaches, mass density fluctuations $\delta\rho$ are related to the Mach number $Ma = |\mathbf{u}|/c_s$, where $\mathbf{u}$ is the flow velocity and c_s is the sound speed, by the scaling $\delta\rho \sim \mathcal{O}(Ma^2)$. This implies that a weakly compressible regime can be achieved in the low-Mach-number limit $Ma \ll 1$. FLAME integrates the compressible formulation of the Hall-MHD equations, although by design, the density fluctuations remain negligible in our simulations. The momentum equation, including the Lorentz force, can be written as

$$\frac{\partial(\rho\mathbf{u})}{\partial t} + \nabla \cdot \left[(\rho\mathbf{u}) \otimes \mathbf{u} + p\mathbf{I} + \frac{1}{2}|\mathbf{B}|^2\mathbf{I} - \mathbf{B} \otimes \mathbf{B} \right] = \rho\nu\nabla^2\mathbf{u}; \quad (1)$$

$$\frac{\partial\rho}{\partial t} + \nabla \cdot (\rho\mathbf{u}) = 0, \quad (2)$$

where ρ is the macroscopic mass density, p is the pressure, $\mathbf{B}$ is the magnetic field, $\mathbf{I}$ is the identity matrix, $\otimes$ denotes the tensor (outer) product between vectors, and ν is the kinematic viscosity. In Eq. (1), the Lorentz force, defined as the cross product between the electric current density $\mathbf{j}$ and the magnetic field, $\mathbf{j} \times \mathbf{B} = (\nabla \times \mathbf{B}) \times \mathbf{B}$, is expressed in a conservative form through the divergence of the Maxwell stress tensor,

$$M_{\alpha\beta} = \frac{1}{2}|\mathbf{B}|^2\delta_{\alpha\beta} - B_\alpha B_\beta, \quad (3)$$

where $\delta_{\alpha\beta}$ is the Kronecker delta, and α and β are the tensor indices of the coordinate system. Compressibility requires the introduction of an equation of state connecting pressure, mass density, and temperature. In this lattice Boltzmann approach, in the low Mach number regime, an isothermal relation $p = c_s^2\rho$ is chosen. Once Eqs. (1) and (2) are

coupled with the Maxwell–Faraday equation for the magnetic field, the system is written as

$$\frac{\partial \mathbf{B}}{\partial t} + \nabla \cdot [(\mathbf{u} - \alpha_H \mathbf{j}) \otimes \mathbf{B} - \mathbf{B} \otimes (\mathbf{u} - \alpha_H \mathbf{j})] = \eta \nabla^2 \mathbf{B}, \quad (4)$$

where η is the magnetic resistivity, and $\alpha_H = L_H/L_0 = v_A/(L_0 u_0) \sqrt{m/(ne^2)}$ is the Hall parameter. Here, v_A is the Alfvén speed, u_0 is the typical fluid velocity, m is the proton mass, n is the proton density, e is the unit electric charge, and L_0 and L_H are the integral length scale (here considered as the box size $L_0 = 2\pi$) and the typical Hall length scale, respectively. We note that if $u_0 = v_A$, L_H matches the proton inertial length d_p . The scale d_p thus varies across the three simulations performed here as a consequence of the chosen α_H value. Indeed, as the box size L_0 is constant, changing α_H modifies d_p , allowing to explore different Hall-MHD turbulence regimes, in which the scale separation between ion and electron dynamics is also changing. Each of the congruent edges of the three-dimensional spatial cube is $L_0 = 2\pi$, discretized with $N = 512$ grid points. As a consequence, the numerical resolution is $\Delta x = \Delta y = \Delta z \sim 0.0123$ for the three runs, which differ for the fluid Reynolds number $Re = U_{rms} L_0 / \nu$ (computed at the time of maximum magnetic dissipation), and for the α_H parameter.⁴⁴ The Prandtl number for all the runs is set to be equal to unity at $t = 0$. During the evolution of the three simulations, however, there can be deviations from the prescribed configuration due to intrinsic differences of the integration schemes for the fluid and magnetic lattices in FLAME (see Foldes *et al.*⁴⁴). In particular, some discrepancy between numerical dissipation of the two schemes is needed in order to attain high Reynolds numbers while maintaining the code stable, also in the presence of fast small-scale phenomena, such as the whistler waves emerging in the Hall regime. We summarize in Table I the parameters characterizing the three runs under study, including the resolution constraints for both the velocity ($k_{max} \eta_K > 1$) and the magnetic ($k_{max} \eta_m > 1$) field fluctuations, $\eta_K = (\nu^3/\varepsilon_u)^{1/4}$ and $\eta_m = (\eta^3/\varepsilon_B)^{1/4}$ being the Kolmogorov viscous and resistive length scales, respectively (here $\varepsilon_u = \nu \langle |\nabla \times \mathbf{u}|^2 \rangle$ and $\varepsilon_B = \eta \langle |\mathbf{j}|^2 \rangle$).

The three runs performed with FLAME have all been initialized with a classical Orszag–Tang vortex configuration, where

$$\mathbf{u}(\mathbf{r}, 0) = u_0 [-2 \sin(y), 2 \sin(x), 0], \quad (5)$$

$$\mathbf{B}(\mathbf{r}, 0) = B_0 [-2 \sin(2y) + \sin(z), 2 \sin(x) + \sin(z), \sin(x) + \sin(y)], \quad (6)$$

with $u_0 = B_0 = Ma/\sqrt{3}$ as the amplitude of the initial perturbations.

Simulations evolved up to their respective peak of magnetic dissipation, then were allowed to decay, and the analysis was performed on a single snapshot taken when the dissipation was maximal, and the

TABLE I. Mach number (Ma), fluid Reynolds number (Re), Hall parameter (α_H), and the plasma beta (β_p) for the three-dimensional Hall-MHD numerical simulations of the parametric study. Prandtl number (Pr) is equal to 1 for the three runs.

RUN	$Ma \times 10^{-4}$	Re	α_H	β_p	$k_{max} \eta_K$	$k_{max} \eta_m$
I	7.0	4400	0.0025	1.1	2.0	1.4
II	1.0	5240	0.01	1.1	2.2	1.4
III	0.625	7150	0.025	1.3	2.6	1.4

system reached a fully developed turbulent state, indicating that the statistical properties of the macroscopic fields are essentially independent of the initial conditions. In the left column of Fig. 1, we present the squared current density j^2 map for RUN I (top), RUN II (middle), and RUN III (bottom). The magnetic $E_B(k)$ and kinetic $E_v(k)$ energy spectra are plotted next to the renderings of j^2 , black and blue lines, respectively, the red-dashed line indicating the Kolmogorov power-law spectrum $k^{-5/3}$ is shown as a reference in all the panels.²³ In RUN I, a broad MHD-scale inertial range appears in both the magnetic and velocity spectra, covering about one decade of scales. As the Hall parameter progressively increases (in RUN II and RUN III), the MHD range shrinks, and a power law compatible with Hall-MHD phenomenological prediction $k^{-7/3}$ (e.g., Alexandrova *et al.*⁴⁶) appears on the small-scale end of the magnetic spectra, reaching its maximum extension in RUN III. This is also evident from the inset in the top panel, where ratios between pairs of spectra are shown for different runs. The break occurring at a different wave number (k) can be identified as the departure from a constant ratio, which shows the increasing expansion of the Hall range toward the larger scales as the Hall parameter increases. As expected, there is a decoupling of the plasma proton flow and the magnetic field, with the velocity spectra not following the characteristic Hall-MHD power law trend at large k .

Figure 2 shows two-dimensional magnetic power spectra for run III, computed in the $z = 0$ (left panel), $y = 0$ (middle panel), and $x = 0$ (right panel) planes. Though with some differences, the spectra show isotropy throughout all the scales, in the three cases examined, confirming that the turbulence has evolved toward a fully developed state.

III. INTERMITTENCY AND ENERGY TRANSFER RATE

Intermittency and energy transfer rate are important features of turbulent flows, determining how structures convey energy toward dissipation⁴⁷ and/or toward the large scale, in the presence of inverse or dual cascades.^{48–50} These can be studied by characterizing the scaling properties of the scale-dependent two-point increments of velocity and magnetic field, defined for a generic vector $\mathbf{A}$ (with $\mathbf{A}$ representing either the current density $\mathbf{j}$, the velocity field $\mathbf{u}$, or the magnetic field $\mathbf{B}$), as $\Delta \mathbf{A} = \mathbf{A}(\mathbf{r} + \mathbf{l}) - \mathbf{A}(\mathbf{r})$, where $\mathbf{l}$ are the increments between two points in the spatial domain. In this context, it is useful to introduce the definition of longitudinal increments $\Delta A_L = \Delta \mathbf{A} \cdot \boldsymbol{\mu}_l$, with $\boldsymbol{\mu}_l = \mathbf{l}/l$ unit vector, and of the ensemble or the volume average $\langle \dots \rangle$ that will be used in the following.

The probability density functions (PDFs) of the standardized (i.e., normalized to the standard deviation) longitudinal increments ΔB_L (Fig. 3) and Δu_L (Fig. 4) help us to quantify the different turbulent states of RUN I (left panel), RUN II (middle panel), and RUN III (right panel), respectively. In both figures, increment scales are expressed in units of grid size. The conversion in physical units can be inferred by normalizing lags in terms of grid points to the proton inertial length (d_p). In each panel, the curves indicate different scale separations l (with red, yellow, and black full lines corresponding to $l \sim 2^7 \Delta x$, $l \sim 2^4 \Delta x$, and $l \sim 2^2 \Delta x$, respectively, for one horizontal component), computed along the Cartesian directions of each field component. This choice ensures that the increments represent longitudinal fluctuations, i.e., variations measured along the direction of the field itself. For comparison, a Gaussian distribution (black dashed lines) is reported in all panels. As expected in the presence of isotropic turbulence, the longitudinal increments of the three components of the fields exhibit consistent statistical properties along the three directions so here we

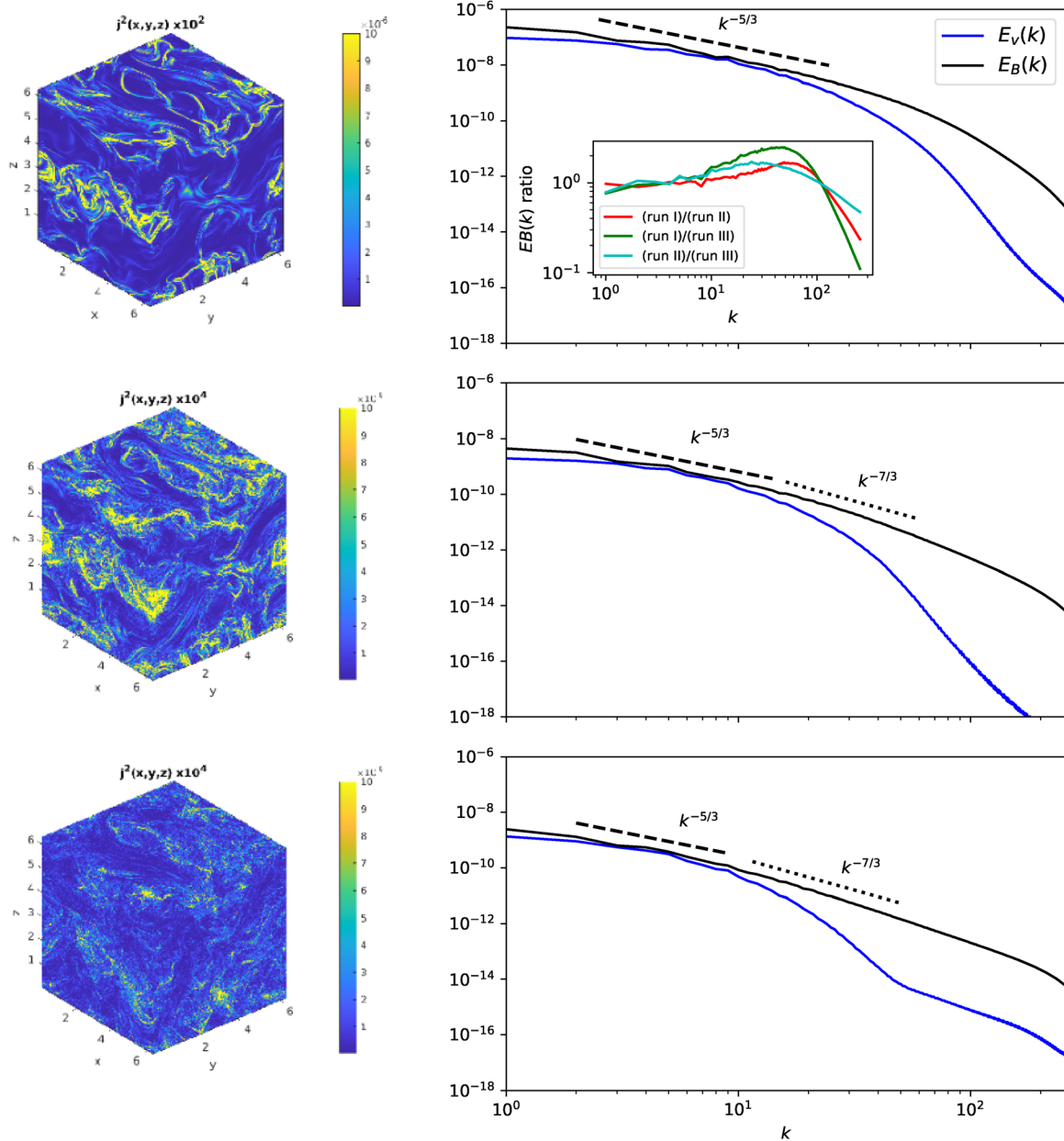


FIG. 1. Left column: current density magnitude $j^2(x, y, z)$. Right column: magnetic (black lines) and kinetic (blue lines) energy spectra. Dashed lines indicate $k^{-5/3}$ and $k^{-7/3}$ references. The final states of RUN I, RUN II, and RUN III are, correspondingly, presented in the first, second, and third rows. The inset in the top panel shows the ratios between magnetic spectra.

computed statistics on ensembles of the three longitudinal increments along Cartesian axes, performed after separately standardizing each component (i.e., fluctuations of the fields, after subtracting their mean, are normalized by their respective standard deviations.). The change in shape of the PDF with the increment scale, evident from Figs. 3 and 4, is the typical signature of intermittency, highlighting the emergence of large-amplitude small-scale fluctuations, inhomogeneously distributed in regions of space where most of the energy dissipation is

concentrated. It is interesting to note that tails in the PDF of the magnetic field in RUN I are clearly higher, for the smaller visualized scales, compared to RUN II and III. This is in agreement with the spectral observation of a broader inertial range, consistent with the intermittent concentration of energy dissipation. This, instead, is not clear when comparing the PDFs of Δu , suggesting that the Hall effect may play a role in modulating small-scale intermittency developing in the magnetic field while decoupling the two fields. In addition to this

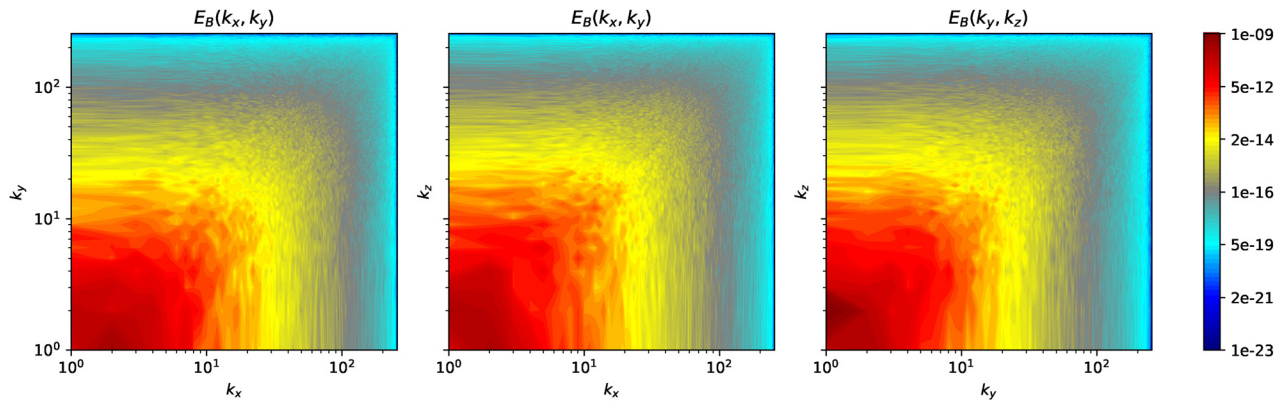


FIG. 2. 2D power spectra of magnetic energy $E_B(k_x, k_y)$ computed at planes $z = 0$ (left panel), $y = 0$ (middle panel), and $x = 0$ (right panel) from RUN III.

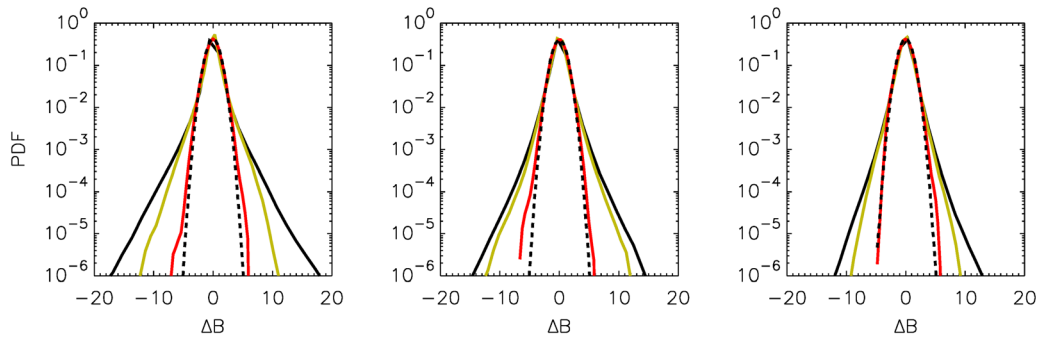


FIG. 3. PDFs of the ΔB standardized values computed for RUN I (left panel), RUN II (middle panel), and RUN III (right panel). We show three different scales: $l \sim 2^2 \Delta x$ (black-solid line), $l \sim 2^4 \Delta x$ (yellow-solid line), and $l \sim 2^7 \Delta x$ (red-solid line). The black-dashed line corresponds to the Gaussian distribution.

physical effect produced by the Hall term, a more conservative integration scheme for the fluid lattice introduces stronger numerical dissipation, as already mentioned in Sec. II, producing a depletion of turbulence fluctuations and intermittency at the very small scales. This numerical effect is more evident the lower the Mach number (going from RUN I to RUN III), as it also appears from the kinetic energy

spectra in Fig. 2, showing exponential cutoffs at smaller k at the smaller Mach values.

An alternative characterization of intermittency can be done through the analysis of the structure functions, defined as the moments of the scale-dependent increments' PDFs, $\langle \Delta v^q \rangle$. In particular, here we use the scaling properties of the flatness F , here computed as the ratio

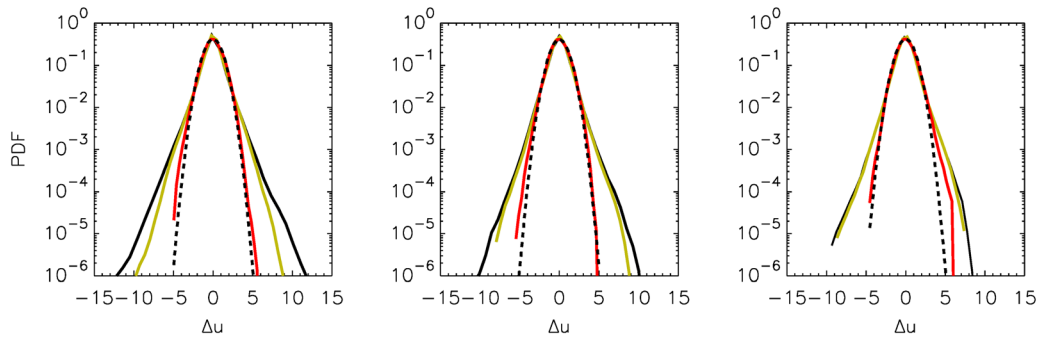


FIG. 4. PDFs of Δu standardized value computed in RUN I (left panel), RUN II (middle panel), and RUN III (right panel). We show three different scales: $l \sim 2^2 \Delta x$ (black-solid line), $l \sim 2^4 \Delta x$ (yellow-solid line), and $l \sim 2^7 \Delta x$ (red-solid line). The black-dashed line corresponds to the Gaussian distribution.

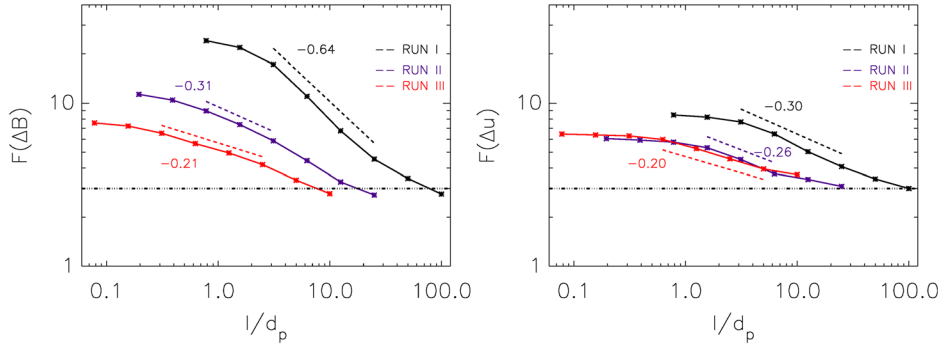


FIG. 5. Flatness F of ΔB (left panel), and Δu (right panel), as a function of scale l/d_p . Black lines correspond to RUN I, blue lines correspond to RUN II, while red lines represent RUN III. A horizontal dashed line refers to $F = 3$.

between the fourth-order and the squared second-order structure functions

$$F(\Delta A) = \frac{\langle \Delta A^4 \rangle}{\langle \Delta A^2 \rangle^2}. \quad (7)$$

The scaling of the flatness of ΔB and Δu (based on the ensemble-averaged longitudinal increments) is presented in the left and right panels of Fig. 5, respectively. In this figure, black lines correspond to RUN I, blue lines to RUN II, and red lines to RUN III. At large scales, the flatness F is close to 3 (horizontal dashed lines) for all runs, indicative of nearly Gaussian statistics. In the MHD inertial range, the flatness increases as the scale decreases, in response to the higher tails emerging in the PDFs of the field fluctuations. This increase is compatible with a power-law scaling, which is a strong signature of the turbulence inertial range. Power-law fits are computed over the range of scales showing approximately a linear trend in the log-log plots, which is different for each run (indicated by dashed lines in the figure). The scaling exponent of the flatness is a reliable quantitative estimator of intermittency. The obtained exponents, ranging between -0.2 and -0.64 , indicate strong intermittency for all runs and are comparable with those observed in solar wind turbulence.^{26,51,52} At small scale, the flatness tends to saturate, consistent with previous analysis performed in kinetic simulations and space plasma observations (e.g., Refs. 53–56). This behavior suggests that nonlinear interactions no longer generate smaller-scale structures, possibly due to dispersive Hall effects⁵⁷ and/or dissipation, as also discussed by Rodríguez Imazio *et al.*⁵⁸ From a comparison of the three runs, we can conclude that intermittency is higher in RUN I, corroborated by the observation of a more developed inertial range. The smaller values of flatness attained for Δu , as well as the less extended linear scaling, resulting in a saturation of F at larger scales, further indicate that turbulence is less developed in the velocity field, and further investigation will be needed to carefully isolate the role of the Hall parameter in modulating the level of intermittency in the plasma velocity. It is worth noting that seeking observational evidence of Hall-scale intermittency for the velocity field⁵⁷ in the solar wind is challenging due to the resolution constraints of spacecraft plasma instruments. They exist, however, for plasmas in the terrestrial magnetosheath.^{35,59}

The Politano–Pouquet law (P&P)^{29,60} is a proportionality relation between the mixed third-order moment of the longitudinal increments of velocity and magnetic field and the increment scale. It has been widely utilized to characterize the energy transfer rate of the magnetohydrodynamic turbulent cascade in the solar wind.²² Generalization of

the P&P law to the Hall-MHD case is due to, e.g., Galtier,³¹ Banerjee *et al.*,⁶¹ Hellinger *et al.*,⁶² and Ferrand *et al.*⁶³ Their approach implies that the energy transfer rate, ε , is constant across the turbulent inertial range, and that ε is roughly equal to the total energy dissipation. The P&P law, including the Hall effects, reads as

$$-4\varepsilon(l) = -\left\langle \nabla \cdot \left[(\Delta \mathbf{u} \cdot \Delta \mathbf{u} + \Delta \mathbf{B} \cdot \Delta \mathbf{B}) \Delta \mathbf{u} - 2(\Delta \mathbf{u} \cdot \Delta \mathbf{B}) \Delta \mathbf{B} - \frac{d_p}{2} (\Delta \mathbf{B} \cdot \Delta \mathbf{B}) \Delta \mathbf{j} + d_p (\Delta \mathbf{B} \cdot \Delta \mathbf{j}) \Delta \mathbf{B} \right] \right\rangle. \quad (8)$$

In this relation, it is easy to identify the internal components of the ensemble average $\langle \dots \rangle$ as spatially distributed quantities that represent the local intensity of energy transfer associated with each interaction term. These terms behave like effective densities of energy flux, indicating how much of the corresponding interaction (e.g., velocity self-interaction, magnetic field interaction, Hall current effects) is available to flow inward or outward across a region of space.

Considering that $\langle \nabla \cdot (\dots) \rangle = -\nabla_1 \cdot \langle (\dots) \rangle$, the integration along the longitudinal displacement, μ_l , in a 3D physical space is

$$-\frac{4}{3} \varepsilon l = \langle (\Delta \mathbf{u} \cdot \Delta \mathbf{u} + \Delta \mathbf{B} \cdot \Delta \mathbf{B}) \Delta u_L - 2(\Delta \mathbf{u} \cdot \Delta \mathbf{B}) \Delta B_L \rangle + \left\langle -\frac{d_p}{2} (\Delta \mathbf{B} \cdot \Delta \mathbf{B}) \Delta j_L + d_p (\Delta \mathbf{B} \cdot \Delta \mathbf{j}) \Delta B_L \right\rangle, \quad (9)$$

which can be rewritten in a compact form as

$$-\frac{4}{3} \varepsilon l = \varepsilon_Y + \varepsilon_H, \quad (10)$$

where $\varepsilon_Y = \langle (\Delta \mathbf{u} \cdot \Delta \mathbf{u} + \Delta \mathbf{B} \cdot \Delta \mathbf{B}) \Delta u_L - 2(\Delta \mathbf{u} \cdot \Delta \mathbf{B}) \Delta B_L \rangle$ and $\varepsilon_H = d_p \langle [-0.5(\Delta \mathbf{B} \cdot \Delta \mathbf{B}) \Delta j_L + (\Delta \mathbf{B} \cdot \Delta \mathbf{j}) \Delta B_L] \rangle$ represent the MHD and Hall contributions, respectively. For the isotropic turbulence case, a key assumption enters the derivation of Eq. (9): the integration of the divergence term can be simplified by performing an ensemble average using only increments $\mathbf{l} = (l_x, l_y, l_z)$ along the three Cartesian axes, rather than integrating over the full three-dimensional scale-space volume. In order to make sure that the choice of the integration scheme does not affect the outcome of the ensemble average,⁶⁴ we compared results obtained through a direct trapezoidal integration rule, $\langle (\dots) \rangle = (1/N) \sum (\dots)$, when up to two consecutive points (in each physical direction) were averaged, and a closed five-point Newton-Cotes integration formula, for which the error estimate is $\sim \mathcal{O}(\Delta x^7)$, when more than 2^1 points were averaged. Eventually, no relevant

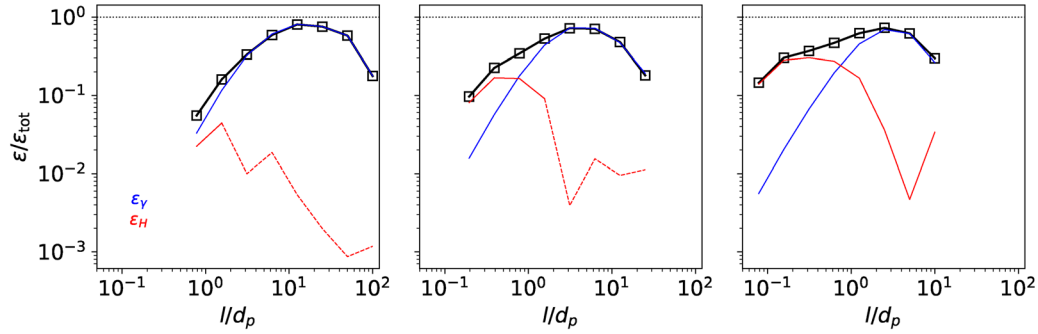


FIG. 6. Energy transfer ε [from Eq. (9)], normalized by the total dissipation, for RUN I (left panel), RUN II (middle panel), and RUN III (right panel). Blue lines identify ε_V and red lines correspond to ε_H .

differences were observed in the calculation of the Hall-P&P law [Eq. (9)] using the two integration schemes (not shown here). Therefore, the results we show are obtained using the trapezoidal integration.

The Hall-P&P law, Eq. (10), is computed for the three runs and presented in Fig. 6 (RUN I, RUN II, and RUN III in the left, middle, and right panels, respectively). For the MHD (blue lines) and Hall (red lines) components, solid and dashed lines indicate positive and negative values, respectively. The dashed lines show the total (kinetic + magnetic) dissipation rate, ε_{tot} , corresponding to the rate of change of the total energy, $-\partial_t E_{tot}$, in decaying turbulence simulations. For the three runs, the values of the total dissipation are 9.8×10^{-13} (RUN I), 4.6×10^{-14} (RUN II), and 6.3×10^{-16} (RUN III). In all cases, the Hall effect appears to be limited, with some positive contributions appearing only in RUN II and III around the Hall scales. The limited statistics and the intrinsic computational challenges introduced by the physics of the Hall term result in a short linear range of the third-order moments. This limitation also stems from the limited number of independent samples considered in the ensemble averages performed. However, the three runs seem to converge accurately to the total dissipation, though in different ranges, with small discrepancies due to the hypotheses required by the P&P law (i.e., homogeneity, isotropy, and negligible dissipation effects in the inertial range), as well as the limited resolution of the simulations. The systematic underestimation of the dissipation rate by the Hall-P&P law reflects the possible violation of its underlying assumptions and the limited scale separation necessary for an ideal inertial range in our simulations. Similar discrepancies have been reported in previous numerical studies (e.g., Adhikari *et al.*⁶⁵ and Wang *et al.*⁶⁶), where full energy balance is recovered only when all terms in the energy evolution equation are included. In our case, finite resolution and domain size, together with numerical dissipation, limit the accuracy of the third-order scaling, particularly at small scales where Hall effects become relevant. Note that, while in all runs, the energy flux is dominated by the MHD terms, the Hall contribution becomes non-negligible at small scales as the Hall parameter increases, as is clear for RUN III (right panel of Fig. 6).

This analysis shows that in the three runs, turbulence is well developed, with the extension of the inertial ranges depending on the Hall parameter, showing strong intermittency, especially in the magnetic field fluctuations. The simulation statistics were indeed sufficient to highlight the emergence of a linear scaling of the third-order law.⁶⁷

IV. LOCAL ENERGY TRANSFER PROXY

The local energy transfer (LET) proxy^{68–70} has been shown to be an effective tool to localize complex turbulence-generated structures in plasmas, such as current sheets and vorticity filaments,^{37,45,71} as well as the energy transfer and conversion associated with their dynamics. The proxy is defined as the point-wise argument of Eq. (9) and is computed at a given scale,

$$\text{LET}(l) \equiv (\Delta \mathbf{u} \cdot \Delta \mathbf{u}) \Delta u_L + (\Delta \mathbf{B} \cdot \Delta \mathbf{B}) \Delta u_L - 2(\Delta \mathbf{u} \cdot \Delta \mathbf{B}) \Delta B_L - \frac{d_p}{2} (\Delta \mathbf{B} \cdot \Delta \mathbf{B}) \Delta j_L + d_p (\Delta \mathbf{B} \cdot \Delta \mathbf{j}) \Delta B_L. \quad (11)$$

Note that the above relation is neither the P&P law (for the global energy transfer rate) nor its weak formulation.^{72,73} It provides a mapping of the scale-by-scale energy flux, emphasizing the energy carried by the local field fluctuations that concur to the global energy cascade. Its different terms also allow to highlight the contribution of vorticity, currents, wave packets (seen as correlated velocity and magnetic fluctuations in the term containing $\Delta \mathbf{u} \cdot \Delta \mathbf{B}$), or Hall currents to the energy transfer process. When the analysis is performed at scales at the bottom end of the inertial range, thus close to the more dissipative scales, the proxy suggests which types of fluctuations are mainly associated with dissipation and where the latter mostly happens in the domain flow, ultimately helping to identify the dissipative processes activated in turbulent plasmas, even in the absence of collisions, such as in space plasmas.¹⁵ To examine the spatial distribution of the turbulent energy transfer in our simulations, we computed the LET proxy, Eq. (11), for various scales, defined as $2^n \Delta x$ (n being an integer), and for the three runs. In Fig. 7, we present three-dimensional maps of the LET at three different scales, namely, $l \sim 2^2 \Delta x$ (left panel), $l \sim 2^4 \Delta x$ (middle panel), and $l \sim 2^7 \Delta x$ (right panel) for RUN III. The transition from a relatively homogeneous distribution of intense regions of energy transfer at large scales to a highly concentrated inhomogeneous distribution at small scales is evident. It can also be appreciated how the flux is characterized pretty much everywhere by alternating negative and positive transfer, respectively associated with energy flux from larger to smaller scale and vice versa. The global flux being negative for a direct cascade,⁶⁰ this visualization highlights the small imbalance between the flux directions, resulting in an overall forward turbulent cascade as expected in three-dimensional turbulence, in the absence of anisotropy. Similar maps for RUN I and II (not shown) do not exhibit major differences.

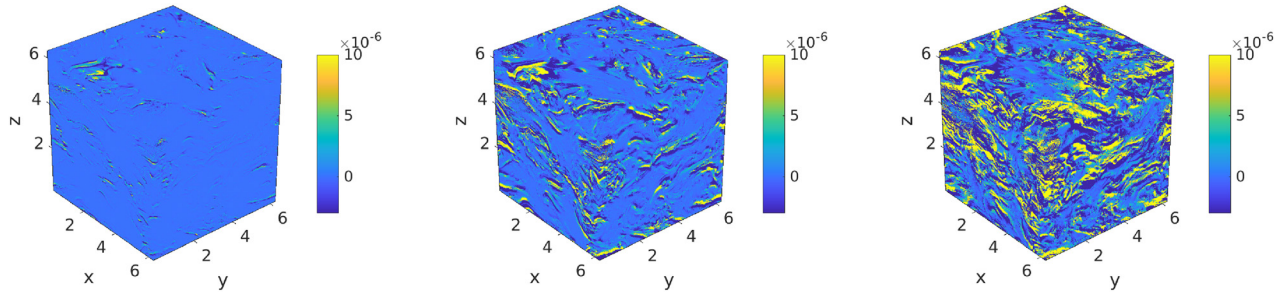


FIG. 7. LET proxy computed in Run III at three different scales. Left panel: $l \sim 2^2 \Delta x$, middle panel: $l \sim 2^4 \Delta x$, and right panel: $l \sim 2^7 \Delta x$.

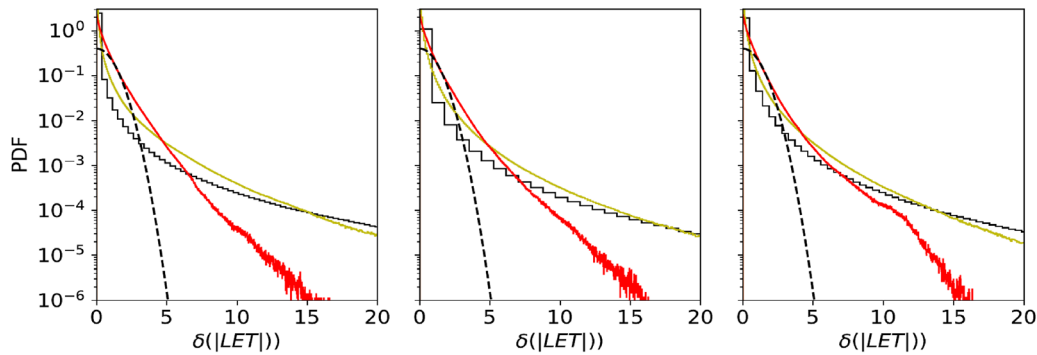


FIG. 8. PDFs of standardized LET proxy ($\delta|LET|$) computed at the scales $l \sim 2^2 \Delta x$ (black-solid line), $l \sim 2^4 \Delta x$ (yellow-solid line), and $l \sim 2^7 \Delta x$ (red-solid line). Left panel: Run I. Middle panel: RUN II. Right panel: RUN III. Absolute values of the proxy were used to improve information clarity. The black dashed line is the positive part of a reference Gaussian distribution.

For a more quantitative comparison between the three runs, we examined the statistical properties of the LET. Figure 8 shows the PDFs of the standardized $\delta|LET|$ proxy, computed at the same scales l , as in Fig. 7. Note that these PDFs have heavier tails than those for the fields' increments, since they refer to a triple product of intermittent quantities. Small differences appear transitioning from RUN I (left panel) to RUN III (right panel). To highlight them quantitatively, in Fig. 9, we also show the flatness of $\delta|LET|$, which reaches extremely large values (up to 10^4) and increases with decreasing scale, following power laws with large exponents between -1 and -2 . These

departures from the Gaussian reference arise because the LET proxy involves third-order mixed products of intermittent quantities, amplifying the contribution of intense localized fluctuations and producing heavy-tailed distributions. Similar behavior has been reported in previous LET-based analyses of simulations and space plasma observations (e.g., Sorriso-Valvo *et al.*⁷⁰ and Hernández *et al.*⁷¹), confirming that such large flatness values are expected. These studies also emphasized that the LET proxy, while not an exact law, is particularly sensitive to localized bursts of energy transfer, which are characteristic of intermittent turbulence.

V. CONCLUSIONS AND DISCUSSION

Critical to understand dynamics in the quasi-collisionless solar wind, the characterization of energy transfer and dissipation across magnetohydrodynamics and kinetic regimes in magnetized turbulent plasmas remains a major challenge.^{8,74} Direct numerical simulations represent a very useful tool for studying dynamics of large-scale structures, waves, and reconnection events as encountered in space plasmas.^{75,76}

In this work, we focus on the study of Hall-MHD plasmas, aiming at characterizing statistics of the field fluctuations, intermittency properties, as well as global and local third-order scaling laws in a regime of interest for the comparison with space observations. Understanding how turbulence develops under different plasma conditions (in this case, varying the contribution of the Hall term) may help to achieve a better understanding of the phenomena connecting

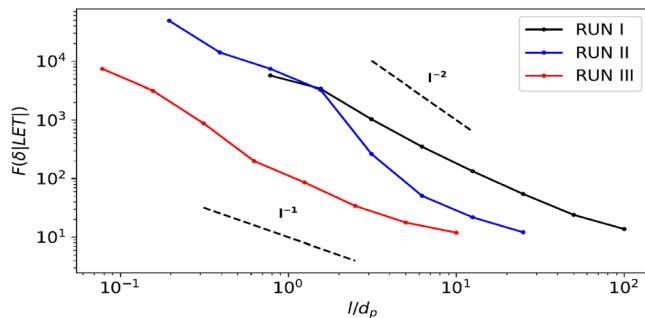


FIG. 9. Flatness F of $\delta|LET|$ at a given scale l/d_p . Black lines correspond to RUN I, blue lines correspond to RUN II, and red lines represent RUN III.

magnetohydrodynamic turbulence and intermittent kinetic processes developing at smaller scales.¹ Here, we studied three different Hall-MHD plasma configurations, all initialized with an Orszag–Tang vortex. The governing Eqs. (1)–(3), which include the Hall-parameter α_H , were numerically solved using the lattice-Boltzmann code FLAME,⁴⁴ as described in Sec. II. Current density and magnetic energy spectra show different yet qualitatively comparable features for an increasing strength of the Hall term, going from $\alpha_H = 0.0025$ (RUN I) to $\alpha_H = 0.025$ (RUN III). For a reduced Hall effect, the MHD inertial range is broader, corresponding to a more developed turbulent state, thus broader Kolmogorov-like spectra. The emerging enhanced intermittency, appreciable through the change of shape of the PDFs of the field increments and of the associated flatness, confirms this evidence. Such a scenario is further supported by the observation of a better-defined linear scaling of the third-order Politano and Pouquet law [Eq. (8)]. As the Hall effect becomes more prominent at larger scales in the simulated plasma (by increasing the parameter α), the MHD range shrinks, and a Hall-MHD range becomes more visible, as also suggested by the steepening of the spectra. However, intermittency is reduced, since the Hall regime does not support a further reduction in the size of turbulent structures.⁵⁷ The range of scales over which the Politano and Pouquet law is observed also reduces, consistently with the reduction of the associated power law range in the spectra, since the Hall contribution seems not to support the energy transfer. A limited role of Hall terms due to the finite size of the domain and/or the limited resolution in numerical simulations has already been previously observed,⁶² while in space plasmas, the Hall contributions appear as more relevant in extending the linear third-order scaling to the Hall range.³⁵ On the other hand, when examining the local properties of the energy transfer using the LET proxy, differences between the three runs reduce considerably, suggesting that the plasma in all simulations is in a state of developed turbulence, with nonlinear interactions being strong at all scales. This preliminary analysis demonstrates the capability and efficacy of the LB approach to investigate the Hall-MHD turbulence, which shows here the expected properties in a series of standard diagnostics conducted on simulations performed with the code FLAME.⁴⁴ Finally, the LET proxy gives a hint of potential uses of these simulations for the investigation of the connections of turbulence-generated structures with dissipation processes. Work in progress aims to compare various definitions of local energy transfer, to study the role of anisotropy at the Hall scales, and, no less important, the influence of numerical dissipation in these types of simulations.

ACKNOWLEDGMENTS

C.L.V. was partially supported by EPN internal Project Nos. PIS-22-01, PIS-22-04, and PIM-23-01, and the CEDIA Project No. PIEX-CEDIA-23-31. L.S.-V. acknowledges the support of the Swedish Research Council (VR) Research Grant No. 2022-03352 and of the PRIN 2022 project “2022KL38BK - The ULtimate fate of TuRbulence from space to laboratory plAsmas (ULTRA)” (Master CUP B53D23004850006) by the Italian Ministry of University and Research, funded under the National Recovery and Resilience Plan (NRRP), Mission 4 – Component C2 – Investment 1.1, “Fondo per il Programma Nazionale di Ricerca e Progetti di Rilevante Interesse Nazionale (PRIN 2022)” (PE9) by the European Union – NextGenerationEU. R.M. and R.F. acknowledge the support from the project “EVENTFUL” (ANR-20-CE30-0011), funded by the French

“Agence Nationale de la Recherche” – ANR through the program AAPG-2020. The simulations have been performed on HPC facilities at École Centrale de Lyon (PMCS2I) in Ecully (France).

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

C. L. Vásconez: Conceptualization (equal); Formal analysis (equal); Supervision (equal); Visualization (equal); Writing – original draft (equal). **R. Földes:** Data curation (equal); Resources (equal); Visualization (equal); Writing – original draft (equal). **R. Marino:** Conceptualization (equal); Formal analysis (equal); Validation (equal); Writing – original draft (equal). **R. E. Manobanda:** Methodology (equal); Resources (equal); Software (equal); Writing – original draft (equal). **E. Leveque:** Resources (equal); Writing – original draft (equal). **D. Telloni:** Resources (equal); Writing – original draft (equal). **L. Sorriso-Valvo:** Conceptualization (equal); Formal analysis (equal); Methodology (equal); Supervision (equal); Writing – original draft (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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